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EE 545 — Theoretical Final Exam Prediction (Calibrated)

Calibration anchor (actual exam, 3 theoretical questions): 1. Orthogonal projections + projection-matrix properties ($P^2 = P$, $P^* = P$, range/null-space meaning, why they work). 2. A defined Hermitian PSD — conceptual/algebraic consequences of A^* , $A^\top$, conjugation, the Hermitian and PSD properties, notation traps. 3. Schur theorem — statement, meaning, proof idea, why every square complex matrix is unitarily triangularizable.

Inferred selection rule (the key sentence): > *The professor asks definitions plus consequences of special matrix structures, especially where notation and proof properties matter.*

Operational decoding of his style: - He picks the **central, repeatedly-motivated structured matrices** (projection, unitary, Hermitian, normal, PD/PSD, Schur/SVD), never peripheral computation. - Every question is **definition + immediate consequence/property + “why”**, almost no arithmetic. - He **rotates among the structured-matrix family**: the three tested archetypes were (A) *structured matrix + properties + why*, (B) *Hermitian/PSD notation + algebraic consequence*, (C) *named theorem: statement + proof idea + meaning*. A future final will reuse these archetypes with **different members of the family**. - He **weaponizes notation**: $A^\top$ vs A^* , real vs complex, Hermitian vs symmetric, PSD vs PD, “diagonalizable” vs “unitarily diagonalizable”, “triangularizable” vs “diagonalizable”. - He likes the **cross-lecture chain**: inner product $\rightarrow$ Cauchy–Schwarz $\rightarrow$ orthogonality $\rightarrow$ projection $\rightarrow$ unitary $\rightarrow$ Hermitian/normal $\rightarrow$ Schur/spectral $\rightarrow$ SVD.

RANKED PREDICTED QUESTIONS (most $\rightarrow$ least likely)

Q1. Normal matrices & the spectral theorem — ~80%

Likely wording. “Define a normal matrix. Prove that a square complex matrix is **unitarily diagonalizable if and only if it is normal**. Explain how unitary, Hermitian, and skew-Hermitian matrices arise as special cases according to where their eigenvalues lie.”

Why he asks it. It is the *direct successor* to the Schur question he already set (Schur $\rightarrow$ normal $\rightarrow$ spectral is his explicit storyline). It is a named theorem with a short, elegant proof built on Schur, plus a “special cases by eigenvalue location” part that tests notation (real axis vs unit circle vs imaginary axis). Maximum concept, minimum calculation.

Lecture patterns. L11–L12 (normal = $A^*A = AA^*$, unitary diagonalization, eigenspace orthogonality); L14/L16 reviews repeatedly draw the “normal family tree.”

Answer outline. - Def: A normal iff $A^*A = AA^*$. - Start from **Schur** $A = UTU^*$, T upper triangular. Normality is preserved by unitary similarity, so T is normal. - Show a **normal triangular matrix is diagonal**: compare the $(1,1)$ entries of T^*T and $TT^* \Rightarrow$ off-diagonal entries of the first row vanish; induct $\Rightarrow T = \Lambda$. Hence $A = U\Lambda U^*$. - Converse: if $A = U\Lambda U^*$ then $A^*A = U|\Lambda|^2 U^* = AA^*$. - Special cases: eigenvalues **real** $\Rightarrow$ **Hermitian**, on the **unit circle** $\Rightarrow$ **unitary**, on the **imaginary axis** $\Rightarrow$ **skew-Hermitian**; distinct-eigenvalue eigenspaces orthogonal.

Marks. 18–20.

Common mistakes. Asserting *all* matrices are unitarily diagonalizable; conflating “diagonalizable” with “unitarily diagonalizable”; skipping the triangular-normal $\Rightarrow$ diagonal step (the heart of the proof); forgetting eigenvalues are complex in general; saying normal $\Rightarrow$ Hermitian.

Q2. A^*A is Hermitian PSD; PD vs PSD — ~75%

Likely wording. “Let A be an arbitrary $m \times n$ **complex** matrix. Prove that A^*A is Hermitian and positive semidefinite for *every* A . State precisely the condition under which A^*A is positive **definite**. Give both the eigenvalue and the quadratic-form definitions of PD and PSD and explain the difference.”

Why he asks it. This is *exactly* his Exam-Question-2 archetype (Hermitian PSD, A^* vs $A^\top$, notation traps) but pivoted to the PD/PSD distinction — his favorite danger topic. It rewards the quadratic-form definition and punishes elementwise misreadings of “ $A \geq 0$ ”.

Lecture patterns. L12–L15: $x^*(A^*A)x = \|Ax\|^2 \geq 0$; PD iff full column rank; “ $A > 0$ is not an entrywise statement.”

Answer outline. - Hermitian: $(A^*A)^* = A^*A$. - PSD: $x^*(A^*A)x = (Ax)^*(Ax) = \|Ax\|_2^2 \geq 0$ for all x . - PD iff $Ax = 0 \Rightarrow x = 0$, i.e. A **has full column rank** (trivial null space); then $x \neq 0 \Rightarrow \|Ax\|^2 > 0$. - Definitions: PD $\Leftrightarrow$ all eigenvalues $> 0 \Leftrightarrow x^*Ax > 0 \forall x \neq 0$; PSD $\Leftrightarrow$ eigenvalues $\geq 0 \Leftrightarrow x^*Ax \geq 0$. PSD allows **zero eigenvalues** / **flat directions**. - Notation trap: “ $A > 0$ ” / “ $A \succ 0$ ” means definiteness, **not** $a_{ij} > 0$.

Marks. 15–18.

Common mistakes. Using $A^\top$ for complex A ; proving only Hermitian and forgetting PSD (or vice versa); claiming A^*A is always PD (true only for full column rank); reading $A \geq 0$ entrywise; confusing “no negative eigenvalues” with “all positive.”

Q3. Unitary matrices: norm/inner-product preservation & eigenvalues — ~65%

Likely wording. “Define a unitary matrix. Prove that multiplication by a unitary matrix **preserves the Euclidean inner product and norm**. What are the possible eigenvalues of a unitary matrix, and why? Contrast a unitary matrix with a Hermitian matrix.”

Why he asks it. Archetype A on a different family member; central to the orthogonality→unitary→Schur chain; tests $U^*U = I$ vs $UU^* = I$, the unit-circle eigenvalue fact, and the unitary-vs-Hermitian danger pair.

Lecture patterns. L10 (preservation properties), L11–L12 (eigenvalues on unit circle, special case of normal).

Answer outline. - Def: $U^*U = UU^* = I$ (square); columns orthonormal. - $\langle Ux, Uy \rangle = (Uy)^*(Ux) = y^*U^*Ux = y^*x = \langle x, y \rangle$; set $y = x \Rightarrow \|Ux\|_2 = \|x\|_2$. - Eigenvalue: $Ux = \lambda x$, $\|x\| = \|Ux\| = |\lambda| \|x\| \Rightarrow |\lambda| = 1$. - Contrast: unitary = eigenvalues on unit circle, $U^{-1} = U^*$; Hermitian = real eigenvalues, $A = A^*$; both are **normal**.

Marks. 12–15.

Common mistakes. Only requiring $U^*U = I$ without noting the square case gives $UU^* = I$ too; proving norm but not inner-product preservation; claiming eigenvalues are real; confusing unitary with orthogonal-projection or Hermitian.

Q4. SVD: statement, existence, four subspaces — ~55%

Likely wording. “State the singular value decomposition of an arbitrary $m \times n$ matrix A . Explain *why it always exists* (proof idea) and how it differs from the eigenvalue decomposition. Show how the SVD provides orthonormal bases for all four fundamental subspaces.”

Why he asks it. “The best thing that happened to us,” heavily motivated; a named decomposition with a conceptual existence argument and a clean subspace payoff — pure concept, no arithmetic. Natural archetype-C question if he rotates past Schur.

Lecture patterns. L21–L22 (geometric sphere→ellipsoid, algebraic $A = U\Sigma V^*$, existence via induced 2-norm + block diagonalization, four subspaces from columns of U, V).

Answer outline. - $A = U\Sigma V^*$, U, V unitary, Σ rectangular diagonal, $\sigma_1 \geq \dots \geq \sigma_r > 0$ real nonneg. - Existence idea: $\sigma_1 = \|A\|_{2,2}$ gives $Av_1 = \sigma_1 u_1$; extend to orthonormal bases; unitary invariance of the 2-norm forces a block-diagonal form ($w = 0$ argument); recurse. Always exists (unlike eigen-decomposition). - Differs from EVD: two bases vs one; any matrix vs square; real nonneg vs complex. - Four subspaces: $\{u_1..u_r\}$ =column space, $\{u_{r+1}..u_m\}$ =left null space, $\{v_1..v_r\}$ =row space, $\{v_{r+1}..v_n\}$ =null space.

Marks. 15–20.

Common mistakes. Writing V instead of V^* ; claiming Σ square; saying singular values can be complex/negative; confusing left/right singular vectors with the subspaces; conflating SVD with eigen-decomposition.

Q5. Projection theorem / orthogonality principle (abstract) — ~50%

Likely wording. “State the projection theorem in an inner product space. Prove that the error of the best approximation in a subspace is **orthogonal** to that subspace, and derive the resulting normal equations / Gram-matrix system. Why does an orthonormal basis simplify the solution?”

Why he asks it. He already tested projection *matrices*; the *abstract projection theorem* is the conceptual twin and the climax of L23 (he explicitly flagged the exam boundary “through the projection theorem inclusive”).

Lecture patterns. L9 (orthogonal projection $P = QQ^*$, $P = A(A^*A)^{-1}A^*$); L23 (projection theorem, Gram matrix, orthonormal simplification).

Answer outline. - Minimize $\|x - \hat{x}\|$, $\hat{x} \in \text{span}\{p_i\}$. Optimal $\Leftrightarrow \langle x - \hat{x}, p_i \rangle = 0 \ \forall i$ (orthogonality principle); Pythagoras shows any in-subspace error component increases the norm. - Write $\hat{x} = \sum \alpha_j p_j \Rightarrow$ Gram system $G\alpha = b$, $G_{ij} = \langle p_j, p_i \rangle$, $b_i = \langle x, p_i \rangle$. - Orthonormal basis $\Rightarrow G = I \Rightarrow \alpha_i = \langle x, p_i \rangle$, $\hat{x} = \sum \langle x, p_i \rangle p_i$. Least squares = special case.

Marks. 15–18.

Common mistakes. Stating the result without the orthogonality proof; forgetting the Gram matrix must be invertible (independent p_i); for the orthogonal projection matrix forgetting $P^* = P$ (see danger topics).

Q6. Cauchy–Schwarz inequality — ~45%

Likely wording. “State and prove the Cauchy–Schwarz inequality in $\mathbb{C}^n$. Explain how it allows the definition of the angle between two vectors, and when equality holds.”

Why he asks it. Foundational, repeatedly motivated as the gateway to angles/orthogonality; short proof; low calculation.

Lecture patterns. L03; reused in L16/L20 norm-inequality discussions.

Answer outline. $|\langle x, y \rangle| \leq \|x\| \|y\|$. Proof: $0 \leq \|x - \frac{\langle x, y \rangle}{\|y\|^2} y\|^2$ expand; equality iff x, y linearly dependent. Angle: $\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} \in [-1, 1]$.

Marks. 10–12.

Common mistakes. Forgetting the conjugate in the complex inner product; wrong equality condition; circular use of the angle definition.

Q7. Sylvester's law of inertia & $A = SS^*$ — ~40%

Likely wording. “Define the **inertia** of a Hermitian matrix and the relation of ***-congruence**. State Sylvester's law of inertia. Use it to prove that every positive definite matrix can be written $A = SS^*$ for some invertible S .”

Why he asks it. A definition-plus-consequence chain he repeatedly stressed (“review star-congruence, Sylvester, square roots — used again and again”). Notation-sensitive (SBS^* vs similarity $S^{-1}BS$).

Lecture patterns. L14–L15.

Answer outline. Inertia (n_+, n_-, n_0) . $A = SBS^*$, S invertible (not necessarily unitary). Sylvester: *-congruent $\Leftrightarrow$ same inertia. PD has inertia $(n, 0, 0)$ = inertia of $I \Rightarrow$ *-congruent to $I \Rightarrow A = SIS^* = SS^*$.

Marks. 12–15.

Common mistakes. Confusing *-congruence with similarity (eigenvalues vs only signs preserved); thinking S must be unitary; claiming square root is unique.

Q8. Cholesky factorization & the Schur-complement argument — ~40%

Likely wording. “State the Cholesky factorization theorem for a positive definite matrix. Outline the proof, and explain *why the Schur complement $M - vv^*/\alpha$ is positive definite*.”

Why he asks it. A named factorization whose proof hinges on the *concept* (*-congruence preserves inertia), not algebra — exactly his taste.

Lecture patterns. L14–L15.

Answer outline. $A \succ 0 \Rightarrow A = LL^*$, L lower triangular with positive diagonal. Partition, block-eliminate to $SAS^* = \text{diag}(\alpha, M - vv^*/\alpha)$; since SAS^* is *-congruent to $A \succ 0$, it is PD, so each diagonal block (incl. the Schur complement) is PD; recurse.

Marks. 12–15.

Common mistakes. Trying to prove the Schur complement PD directly from the formula (the trap the professor warns against); forgetting the positive-diagonal requirement; confusing with LU.

Q9. Hermitian $\Rightarrow$ real eigenvalues & orthogonal eigenvectors — ~35%

Likely wording. “Prove that the eigenvalues of a Hermitian matrix are real and that eigenvectors corresponding to distinct eigenvalues are orthogonal.”

Why he asks it. Tiny, clean, definition-driven; could appear as a sub-part of the Hermitian/PSD slot.

Answer outline. $Ax = \lambda x \Rightarrow x^*Ax = \lambda\|x\|^2$; $(x^*Ax)^* = x^*A^*x = x^*Ax$ real $\Rightarrow \lambda$ real. For $\lambda \neq \mu$: $\lambda\langle x, y \rangle = \langle Ax, y \rangle = \langle x, Ay \rangle = \mu\langle x, y \rangle \Rightarrow \langle x, y \rangle = 0$.

Marks. 8–12.

Common mistakes. Dropping conjugates; assuming real symmetric only; not using Hermitian-ness in the cross term.

Q10. Four fundamental subspaces & orthogonality relations — ~30%

Likely wording. “Define the four fundamental subspaces of A . State and justify the orthogonality relations $\mathcal{N}(A) \perp \mathcal{R}(A^*)$ and $\mathcal{N}(A^*) \perp \mathcal{R}(A)$. How does the SVD exhibit orthonormal bases for all four?”

Why he asks it. The analysis backbone of the whole $Ax = b$ storyline; conceptual.

Marks. 12–15.

Common mistakes. Mixing up which space pairs with which; forgetting dimensions sum to n (or m); confusing row space with column space.

DANGER TOPICS (simple-looking, high test-probability)

Trap	What students confuse	What to say
A^* vs $A^\top$	use transpose for complex matrices	$A^* = \overline{A}^\top$; they coincide only for real matrices
Hermitian vs symmetric	“symmetric” is the real case	Hermitian $A = A^*$; symmetric $A = A^\top$ (real)
PSD vs PD	“no negatives” vs “all positive”	PSD: $\lambda \geq 0$, $x^*Ax \geq 0$ (flat dirs allowed); PD: $\lambda > 0$, strict
projection vs orthogonal projection	both satisfy $P^2 = P$	oblique projection: $P^2 = P$ only; orthogonal: $P^2 = P$ and $P^* = P$
diagonalizable vs unitarily diagonalizable	general T vs unitary U	unitarily diagonalizable $\Leftrightarrow$ normal ; diagonalizable is weaker
Schur vs diagonalization	triangular always vs diagonal sometimes	Schur $A = UTU^*$ exists for every square A ; diagonalization may fail
unitary vs Hermitian	both “nice” normal matrices	unitary: $ \lambda = 1$, $U^*U = I$; Hermitian: $\lambda \in \mathbb{R}$, $A = A^*$
normal vs Hermitian	Hermitian $\subsetneq$ normal	normal: $A^*A = AA^*$ (eigenvalues anywhere); Hermitian: real eigenvalues
similarity vs *-congruence	$S^{-1}BS$ vs SBS^*	similarity preserves eigenvalues; *-congruence preserves only inertia
“ $A > 0$ ” entrywise vs definite	reads $a_{ij} > 0$	means positive definite, not elementwise

EXAM-STRUCTURE ADJUSTMENT — Final emphasizes L13–23

Fact: the **midterm** covered L1–12; the **final covers everything but emphasizes the later material (L13–23)**. The calibration exam’s three questions (projections L9, Hermitian-PSD L11–14, Schur L11) confirm he *still* pulls the canonical structured matrices whose **roots are early but whose payoff is in L13–23** (normal→spectral, PSD→inertia→Cholesky, Schur→SVD). So the reweighting is: **keep the cross-boundary structured-matrix theorems at the top, raise the pure-L13–23 topics, and lower the pre-L13-only topics** (Cauchy–Schwarz L3, plain unitary L10) since the midterm already owned them.

Rewighted ranking (final, emphasis L13–23):

Rank	Question	Lectures	Prob
1	Normal matrices + spectral theorem (unitarily diag. $\Leftrightarrow$ normal)	L12–13	80%
2	A^*A Hermitian PSD; PD vs PSD; full-column-rank	L13–15	75%
3	SVD: statement, existence, four subspaces	L21–22	70% ↑
4	Inner product spaces + projection theorem (he named it the exam boundary)	L23	68% ↑
5	Schatten / nuclear norm as convex relaxation of rank	L23	55% ↑(new)
6	Sylvester’s law of inertia + $A = SS^*$	L14–15	55% ↑
7	Matrix norms: Frobenius vs operator (σ_1) + unitary invariance	L20	52% ↑(new)
8	Cholesky + Schur-complement-is-PD argument	L14–15	50% ↑
9	Schur complement / block PD criterion / Woodbury	L18	42% (new)
10	Eckart–Young best low-rank approximation	L22	40% (new)
—	Unitary preserves inner product/norm	L10	45% ↓

Rank	Question	Lectures	Prob
—	Hermitian $\Rightarrow$ real eigenvalues / orthogonal eigenvectors	L11–13	38%
—	Cauchy–Schwarz	L3	25% $\downarrow$

New L13–23 questions to add to your prep (high emphasis):

- **Q5’ Schatten/nuclear norm (L23, ~55%).** “Define the Schatten p -norms of a matrix. Show that $p = 2$ gives the Frobenius norm and $p = \infty$ the operator norm. Explain why minimizing the **nuclear norm** ($p = 1$) is a convex relaxation of rank minimization.” Answer: $\|A\|_{(p)} = \|\sigma\|_p$; $p = 1$ is ℓ_1 of singular values $\Rightarrow$ sparsifies $\sigma \Rightarrow$ minimizes $\#\text{nonzero } \sigma = \text{rank}$; convex, SDP-solvable; matrix completion / Netflix. Mistakes: confusing Frobenius with operator norm; thinking rank itself is convex.
- **Q7’ Matrix norms (L20, ~52%).** “Distinguish the Frobenius norm from the induced 2-norm. Prove both are unitarily invariant and express each in terms of singular values.” Answer: $\|A\|_F = \sqrt{\sum \sigma_k^2} = \sqrt{\text{tr } A^*A}$; $\|A\|_2 = \sigma_1$; invariance via trace cyclicity / ratio of invariant norms. **Danger:** $\|A\|_2 \neq \|A\|_F$.
- **Q9’ Schur complement (L18, ~42%).** “For Hermitian $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, state the PD criterion via the Schur complement. Why is $A_{11} \succ 0$ and $A_{22} \succ 0$ not sufficient?” Answer: $A \succ 0 \iff A_{11} \succ 0$ and $S_{11} = A_{22} - A_{21}A_{11}^{-1}A_{12} \succ 0$; subtracting a PSD term can break definiteness.
- **Q10’ Eckart–Young (L22, ~40%).** “State the best rank- p approximation theorem. Why is the rank- p constraint non-convex, yet the SVD gives the global optimum?” Answer: truncate SVD to p terms; error $\sqrt{\sum_{k>p} \sigma_k^2}$ (Frobenius), σ_{p+1} (2-norm); rank set non-convex (sum of two rank-1 can be rank-2) but SVD truncation is the global optimum.

A. TOP 10 MOST LIKELY EXAM QUESTIONS (final, emphasis L13–23 — see reweighted table above)

1. Normal matrices + spectral theorem (unitarily diagonalizable $\Leftrightarrow$ normal) — **80%**
2. A^*A Hermitian PSD; PD vs PSD; full-column-rank condition — **75%**
3. SVD: statement, why it always exists, four subspaces — **70%**
4. Inner product spaces + projection theorem + Gram normal equations — **68%**
5. Schatten / nuclear norm as convex relaxation of rank — **55%**
6. Sylvester’s law of inertia + $A = SS^*$ for PD — **55%**
7. Matrix norms: Frobenius vs operator + unitary invariance — **52%**
8. Cholesky factorization + Schur-complement-PD argument — **50%**
9. Schur complement / block PD criterion / Woodbury — **42%**
10. Eckart–Young best low-rank approximation — **40%**

B. TOP 10 DEFINITIONS TO MEMORIZE EXACTLY

1. **Unitary:** $U^*U = UU^* = I$ (orthonormal columns).
2. **Hermitian:** $A = A^*$ (real eigenvalues); symmetric = real case $A = A^\top$.
3. **Normal:** $A^*A = AA^*$ ($\Leftrightarrow$ unitarily diagonalizable).
4. **Positive definite / semidefinite:** $x^*Ax > 0 / \geq 0$ for all $x (\neq 0)$; equivalently eigenvalues $> 0 / \geq 0$ (Hermitian assumed).
5. **Orthogonal projection matrix:** $P^2 = P$ and $P^* = P$; $P = QQ^*$ (orthonormal Q) or $A(A^*A)^{-1}A^*$.
6. **Schur factorization:** $A = UTU^*$, U unitary, T upper triangular (every square complex A).
7. **SVD:** $A = U\Sigma V^*$, U, V unitary, Σ rectangular diagonal with $\sigma_k \geq 0$.
8. **Inertia / *-congruence:** (n_+, n_-, n_0) ; $A = SBS^*$, S invertible.
9. **Inner product (axioms):** conjugate symmetry, linearity in 1st arg, $\langle x, x \rangle \geq 0$ with $= 0$ iff $x = 0$; induced norm $\sqrt{\langle x, x \rangle}$.
10. **Four fundamental subspaces:** $\mathcal{R}(A), \mathcal{N}(A^*), \mathcal{R}(A^*), \mathcal{N}(A)$ with $\mathcal{N}(A) \perp \mathcal{R}(A^*)$, $\mathcal{N}(A^*) \perp \mathcal{R}(A)$.

C. TOP 10 THEOREM / PROOF IDEAS

1. **Spectral theorem (normal):** Schur $\Rightarrow$ triangular-normal-is-diagonal (compare T^*T, TT^* diagonals).
2. $A^*A \succeq 0$: $x^*A^*Ax = \|Ax\|^2$; PD iff full column rank.
3. **Unitary preserves inner product:** $\langle Ux, Uy \rangle = y^*U^*Ux = \langle x, y \rangle$; $|\lambda| = 1$.
4. **Hermitian real eigenvalues:** x^*Ax real; distinct eigenvectors orthogonal.
5. **Schur existence:** induct by deflating one (unit) eigenvector into an orthonormal basis.
6. **SVD existence:** $\sigma_1 = \|A\|_{2,2}$; orthonormal extension + 2-norm unitary invariance forces $w = 0$; recurse.
7. **Projection / orthogonality principle:** error $\perp$ subspace minimizes distance (Pythagoras) $\Rightarrow$ Gram normal equations.
8. **Sylvester inertia** $\Rightarrow A = SS^*$: PD has inertia $(n, 0, 0)$ = that of I .
9. **Cholesky:** block elimination + *-congruence preserves inertia $\Rightarrow$ Schur complement PD $\Rightarrow$ recurse.
10. **Cauchy-Schwarz:** minimize $\|x - cy\|^2$ over c ; equality iff dependent.

D. MOCK FINAL EXAM (3 major theoretical questions, calibrated style)

Question 1 — Structured matrices & the spectral theorem (35 marks). (a) Define a **normal** matrix and state precisely what “unitarily diagonalizable” means. (5) (b) Prove that a square complex matrix is unitarily diagonalizable **if and only if** it is normal. You may invoke the Schur factorization. (15) (c) For a normal matrix, state where the eigenvalues lie when the matrix is (i) unitary, (ii) Hermitian, (iii) skew-Hermitian, and justify each. (9) (d) True/false with one-line reasons: “Every diagonalizable matrix is normal”; “ $A = A^\top$ implies A is Hermitian.” (6)

Question 2 — Hermitian, PSD, and notation (35 marks). Let A be an arbitrary $m \times n$ **complex** matrix. (a) Show A^*A is Hermitian. State why you must write A^* , not $A^\top$. (6) (b) Prove A^*A is positive **semidefinite** for every A . (8) (c) State and justify the exact condition under which A^*A is positive **definite**. (8) (d) Give the eigenvalue and quadratic-form definitions of PD and PSD and explain the difference; explain why “ $A \succeq 0$ ” is **not** an entrywise statement. (8) (e) If $A \succ 0$ (Hermitian), explain why it can be written $A = SS^*$ and name one such S . (5)

Question 3 — A named theorem: statement, meaning, proof idea (30 marks). (a) State the **Schur factorization** theorem and explain precisely why it differs from diagonalization. (8) (b) Give the **proof idea** (deflation via an orthonormal basis) for why every square complex matrix is unitarily triangularizable. (12) (c) Explain the geometric/conceptual **meaning**: what does “triangular in a unitary basis” tell you, and why is a unitary basis preferable to a general one? (6) (d) Notation check: contrast Schur triangularization with the SVD $A = U\Sigma V^*$ — one basis vs two, square vs rectangular, triangular vs diagonal. (4)

(Marking philosophy mirrors the real exam: credit for correct definitions, correct $/\top$ usage, the key proof step, and the conceptual “why” — minimal credit for computation.)*